



FINAL RECOMMENDATIONS

GROUP 21

LSE DATA ANALYTICS CAREER ACCELERATOR
Employer Project - MSF OC Brussels

21st November 2025



Today, we will cover...

- 1** **Context** - the problem and our approach
- 2** **Findings** - what we learned
- 3** **Recommendations** - what next?



Objective: use insights from the end of assignment (EoA) survey to improve the operational quality of MSF projects

Our objective

Help MSF use insights from the EOA survey to improve quality of operations



The problems it addresses



End of Assignment (EoA) survey response rate is low



Feedback from the survey is not used to inform delivery of projects



Why this matters

MSF has a **duty to provide high quality experience** to the populations it serves and its staff

Method: we transformed the data and ran a series of analyses to inform our dashboard design

1. Import the data

CompletedDate_EndofAssignment	Coordinator_EndofAssignment	DevelopmentAdvisor_EndofAssignment
06 January 2024	Priya Zuberi	Koti Qureshi
06 January 2024	Priya Zuberi	Koti Qureshi
06 January 2024	Priya Zuberi	Koti Qureshi

- Python was used to clean and transform the survey data
- Identified issues in the data to inform technical recommendations

2. Transform & clean it

```
Section 1: Loading packages and setting functions
1.1 Loading packages

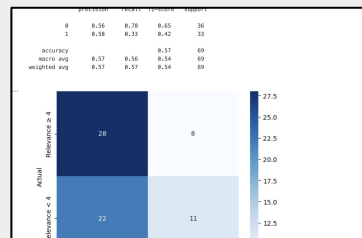
# Import the necessary libraries.
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings

# Predictive analysis libraries
import sklearn.metrics as metrics
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split, cross_val_score, GridSearchCV
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.metrics import accuracy_score, precision_score, recall_score
from sklearn.metrics import roc_auc_score, roc_curve

# Helper: Print all columns in dataset
print_columns = lambda df: print(df.columns)

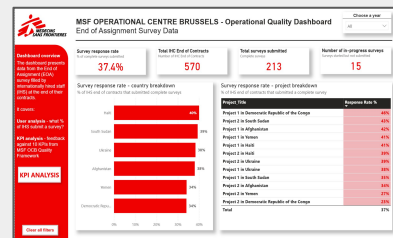
1.2 Set functions
```

3. Run the analysis



- Ran descriptive, exploratory and predictive analysis on the synthetic data

4. Develop the dashboard



- Analysis was used to inform the design and structure of the dashboard

Scope: our analysis focused on how well MSF projects meet local needs and mitigate risks

Objective of our analysis:

1. Understand how high performing MSF projects have delivered impact to local populations
2. Determine what predicts a project's ability to deliver impact to local population

How did we define success?



Relevance - to what extent did projects meet critical needs of the local population



Do No Harm - to what extent did projects identify and mitigate risks for local populations



Note: We didn't use Effectiveness in our analysis because of the way the question was structured in the synthetic data

Context

Findings

Recommendations

Finding 1: Survey response rates are critically low across all projects and countries

Key Takeaway

- Every single country and project is below MSF's target of 50% response rate

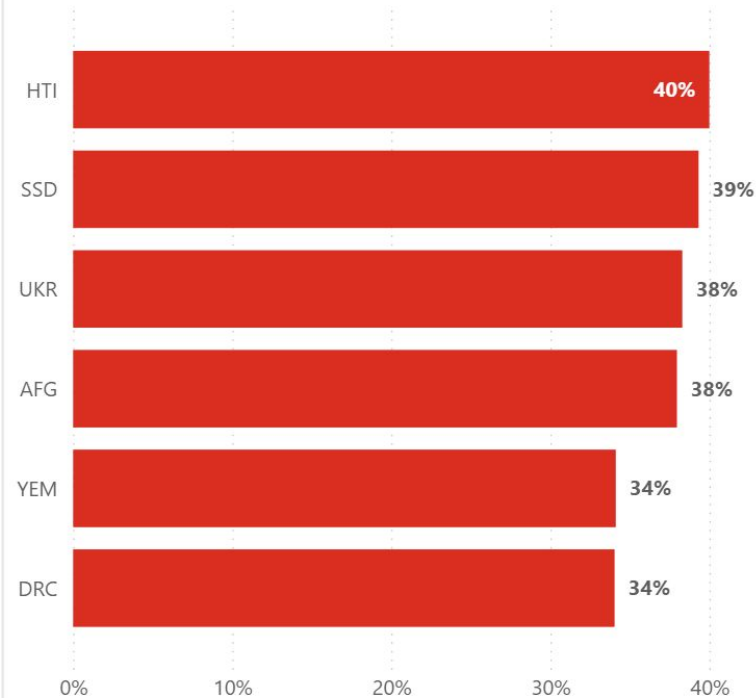
Survey response rate - project breakdown

% of IHS end of contracts that submitted a complete survey

Project Title	Response Rate %
Project 1 in Democratic Republic of the Congo	46%
Project 2 in South Sudan	43%
Project 1 in Afghanistan	42%
Project 1 in Yemen	41%
Project 1 in Haiti	41%
Project 2 in Haiti	39%
Project 2 in Ukraine	39%
Project 1 in Ukraine	38%
Project 1 in South Sudan	35%
Project 2 in Afghanistan	34%
Project 2 in Yemen	27%
Project 2 in Democratic Republic of the Congo	23%
Total	37%

Survey response rate - country breakdown

% of IHS end of contracts that submitted complete surveys



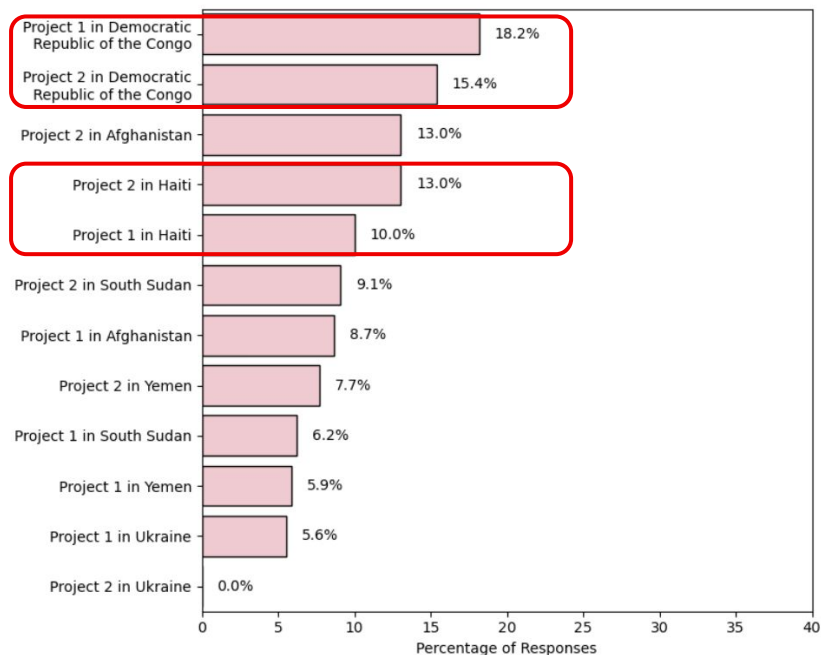
Finding 2: For each KPI, there are some countries who perform poorly across all projects

Key Takeaways

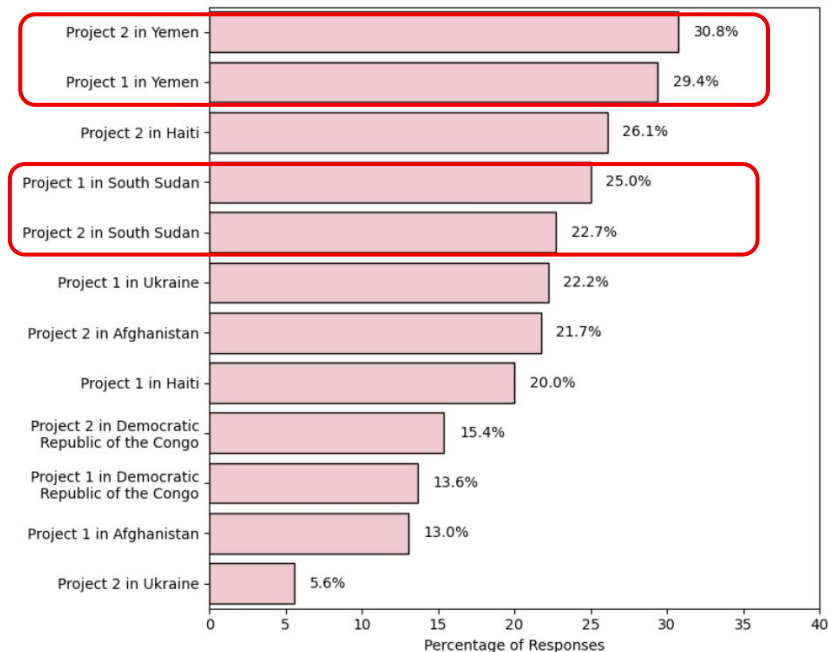
- DRC and Haiti perform poorly across all projects for successfully meeting the needs of the local population

- Yemen and South Sudan perform poorly across all projects for identifying and mitigating risks effectively

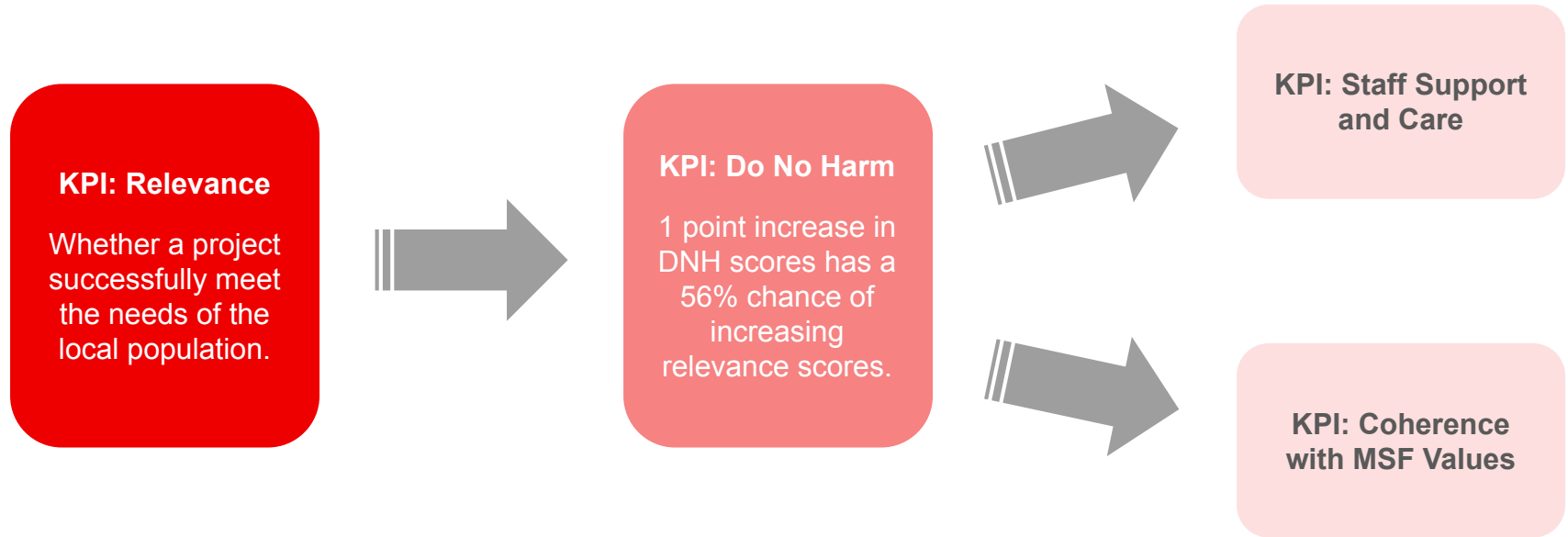
Relevance - % of Low (1-2) Scores



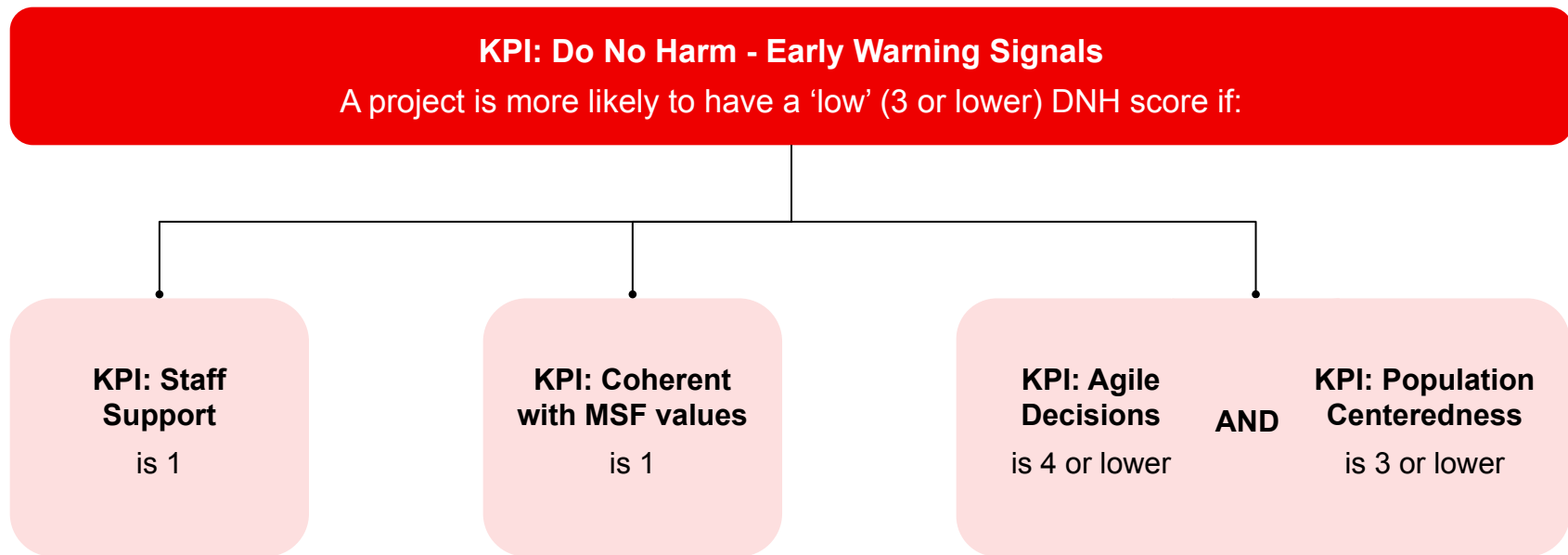
Do No Harm - % of Low (1-2) Scores



Finding 3: There are 'early warning signals' which can be used to identify whether projects will successfully deliver impact (1/2)



Finding 3: There are 'early warning signals' which can be used to identify whether projects will successfully deliver impact (2/2)



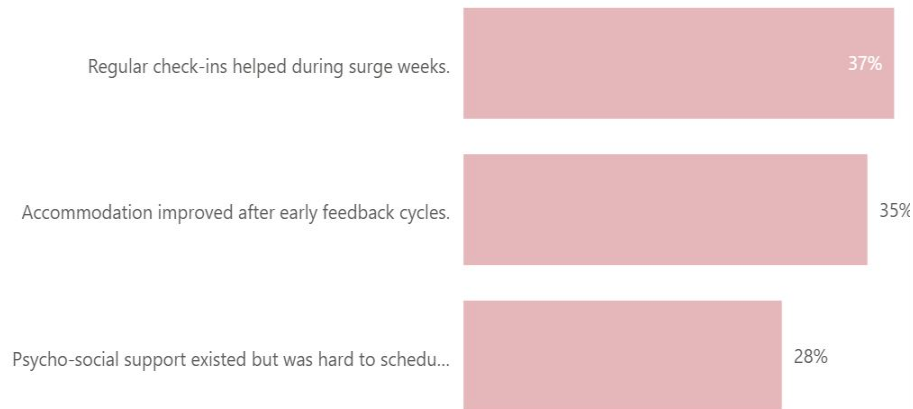
Finding 4: The biggest challenges staff face are psychological support, safety & security, living conditions and workload

Increasing staff support scores by 1 point have a 31% chance of increasing Do No Harm scores.

What are the challenges staff face?

psychological support
training and development opportunities
living conditions
health and medical support
safety and security
workload
communication with management

What staff support was effective?

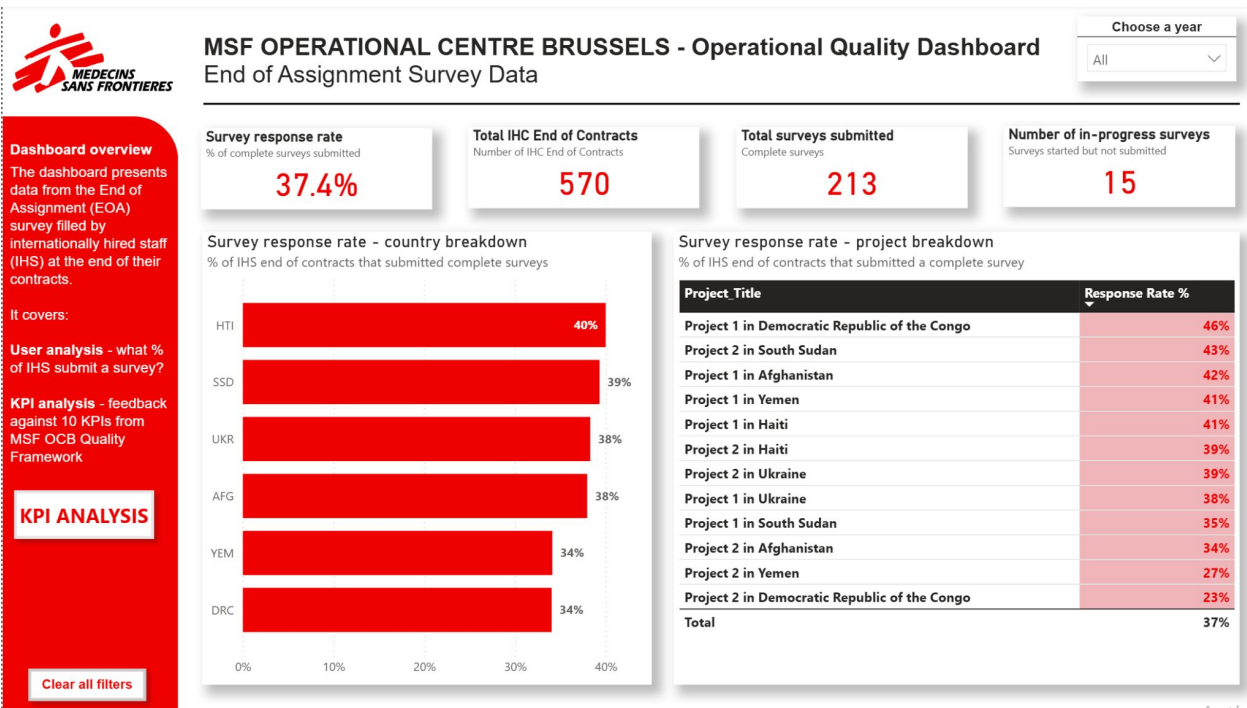


Recommendation 1: Improve survey response rate by using the dashboard and incentivising submissions

Insight: Survey response rates are critically low across all projects and countries.

We recommend:

- **Flag low response countries** and ensure management drive completion.
- **Consider incentivising** survey completion or adding penalties for non-compliance.



Recommendation 2: Use our dashboard warning system to flag 'at risk' projects and intervene early

**MSF OPERATIONAL CENTRE BRUSSELS - Operational Quality Dashboard**
End of Assignment Survey data - KPI Analysis

← Main page

At Risk Tracker - Projects that are at risk of poor performance

Our analysis revealed that the best predictor of whether a project meets the most critical needs of a local population is whether the project identifies and mitigates risk. And the best predictor of whether a project successfully identifies and mitigates risk is whether the project is coherent with MSF values and if staff are supported and cared for. We have created this tracker as an early warning sign that projects are at risk when these three KPIs fall below certain 'thresholds'. Project leads can use this tracker to provide targeted support to projects and proactively improve performance.

Note: The data has been altered to depict the at risk icons.

Choose a year
All

At risk projects - projects below threshold for critical KPIs

Title Projects	Do No Harm	Coherent with MSF Values	Staff is supported and cared for
Project 1 in Afghanistan	⊗	2	2
Project 1 in Democratic Republic of the Congo	⊗	2	2
Project 1 in Haiti	⊗	2	2
Project 1 in South Sudan	⊗	2	2
Project 1 in Ukraine	⊗	2	2
Project 1 in Yemen	⊗	2	2
Project 2 in Afghanistan	⊗	2	2
Project 2 in Democratic Republic of the Congo	⊗	2	2
Project 2 in Haiti	⊗	2	2
Project 2 in South Sudan	⊗	2	2
Project 2 in Ukraine	⊗	2	2
Project 2 in Yemen	⊗	2	2

When projects fall below these thresholds, a **red icon** appears on the dashboard flagging that the project is at risk

Thresholds for a project to be considered at risk:
Do no Harm <2.5
Coherent with MSF values < 2.5
Staff is supported and cared for <1.5

Insight: Process KPIs act as an early warning system for poor project performance.

We recommend:

- **Use our risk tracker** in the dashboard where projects that fall below 'thresholds' can be identified early.
- **Investigate flagged projects** to understand design and implementation issues.

Recommendation 3: Create a holistic system to manage performance and improve the delivery of projects.

Insights

Insight: Projects show clear 'high' and 'low' performance against impact KPIs

Insight: Staff report major challenges around psychological support, safety, and conditions.



Recommendation: A holistic performance management system

Features of a holistic monitoring system:

- 1 A small number of KPIs that are tracked regularly
- 2 Staff that understand the KPIs their performance is being measured against
- 3 A forum where performance is discussed, best practices are shared and learning takes place
- 4 Incentives and penalties that form a robust accountability mechanism

Summary & Conclusion

The EOA survey is a **powerful insight source**, but success doesn't depend on a high quality dashboard **but on how well MSF teams utilise the data to act.**

To strengthen evidence-driven impact, we recommend:

1. **Increase survey completion rates** to improve quality of data
2. **Act early** by using our risk tracker
3. Build a **holistic performance management system**



THANK YOU FOR YOUR TIME

QUESTIONS AND FEEDBACK

LSE DATA ANALYTICS CAREER ACCELERATOR
Employer Project - MSF OC Brussels

14 November 2025



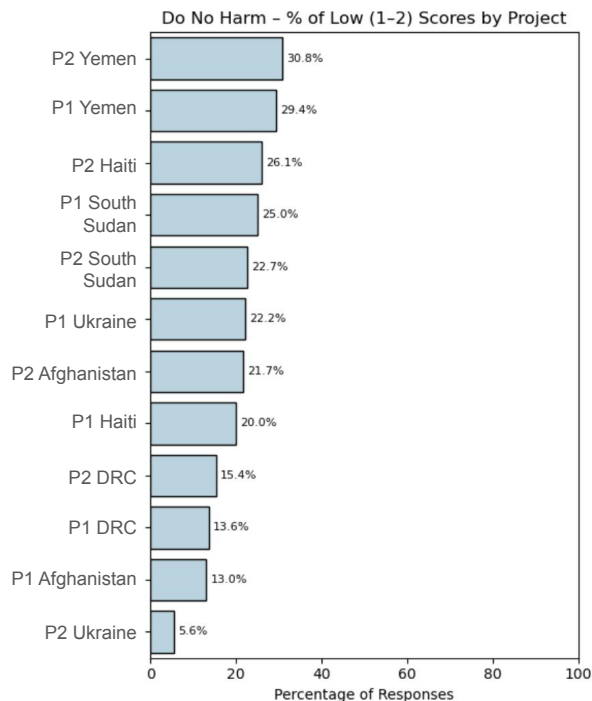
Appendix

We found that the EOA survey provides rich insights that can improve project operations

Afghanistan had the best over all performance



Projects 1 & 2 in Yemen had the highest % of negative scores in DNH



Countries that performed well had a combination of best practices

- 1 Regular check-ins
- 2 Feedback cycles
- 3 Psycho-support
- 4 Community dialogue

Business recommendations - holistic monitoring to drive change

- 1 Importance of data quality** - Introduce incentives or integrate survey in routine processes to encourage high participation.
- 2 Dashboard accessibility** - Consider staff training to understand indicators and its importance in decision making.
- 3 Conduct routine meetings with stakeholders** - Review dashboard regularly and discuss specific challenges in projects to drive improvement.
- 4 Culture of learning and accountability** - Develop case studies and best practices from projects that perform well to drive improvement.

Types of descriptive and exploratory analyses executed



Country level - which countries are high and low performers in relevance and DNH and why?



Project level - which projects are high and low performers in relevance and DNH and why?



Staff level - how does staff respond in different countries and projects?



Qualitative - what can we uncover from people's qualitative responses about relevance and do no harm?

Technical recommendations - survey & KPI improvements

1 Clarify scoring method - Current KPI calculation treats Likert scale data as a continuous variable.
Proposal: Convert survey to continuous scoring method or revise KPI calculation to % of respondents rating 4–5.

2 Use median score over mean - Current KPI score is calculated using mean.
Proposal: Use median to represent average choice as it is not susceptible to extreme values.

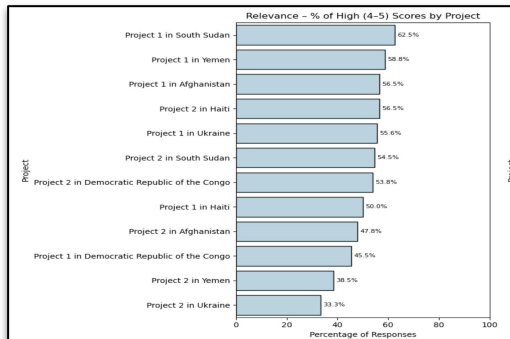
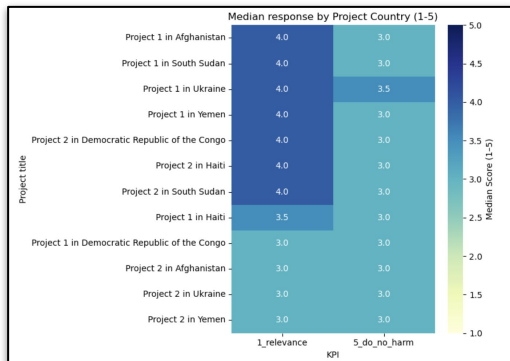
3 Consider a mid-point survey - Current data is submitted only at EOA to inform future assignments.
Proposal: Implement a mid-point check-in or survey to tackle conditions in active projects.

4 Launch a project scorecard - This will allow project leads to easily identify gaps.
Proposal: Implement a scorecard to depict all project performance KPIs and provide insights for improvement.

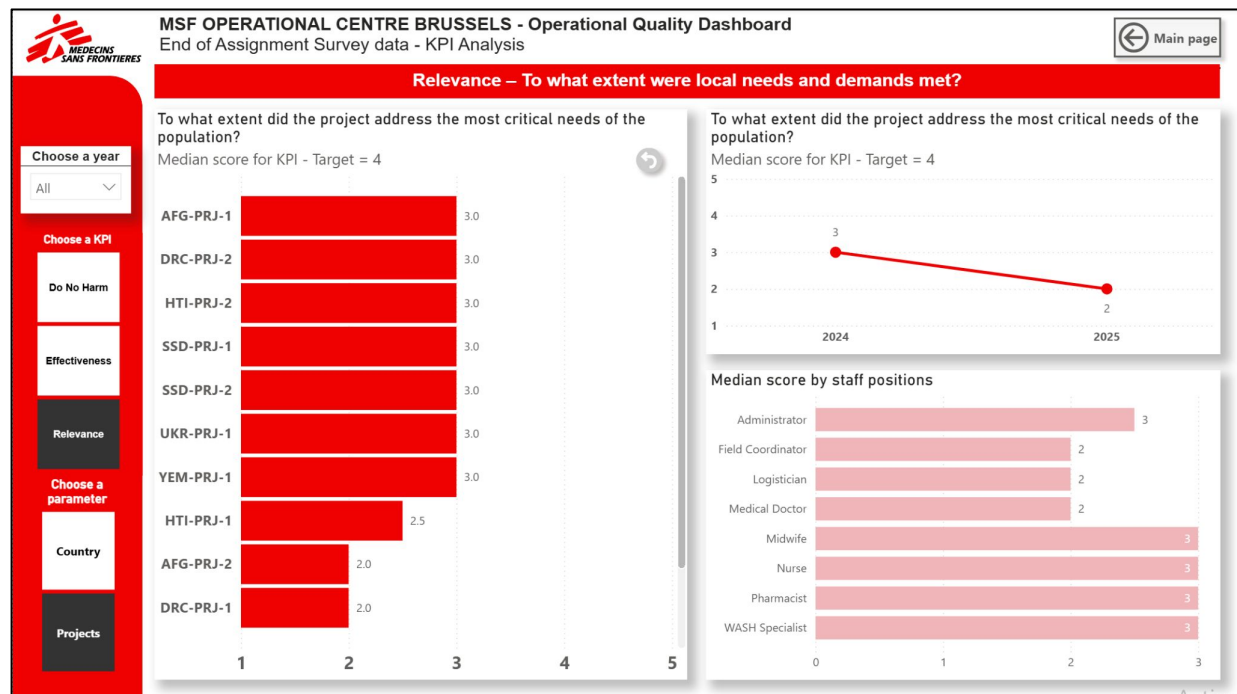
This also allowed us to test various indicators and visuals for our dashboard

Exploration of different KPI indicators like median scores and percentage of respondents that chose high scores (4 & 5) and low scores (1 & 2).

Sample from our analysis



Dashboard extract



2

3

4

Technical Report - Explains technical aspects of analysis, the thought process and analytical decisions taken to design dashboard

1

[illegible]

- MSF OPERATIONAL CENTRE BRUSSELS - Operational Quality Dashboard**

End of Mission Survey Data

Current survey results

Survey response (n = 104)	Total MSF Operational Centre Brussels	Total survey responses	Number of projects included
37.4%	570	213	15

Survey response (n = 104) breakdown

10 of 104 respondents (9.6%) were from the national government and 94 (90.4%) were from other stakeholders.

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
Government (n = 10)	100%
Non-governmental (n = 94)	100%
Academic (n = 0)	0%
Private sector (n = 0)	0%
Other (n = 0)	0%

Survey response (n = 104) by stakeholder type

Stakeholder type	Survey response (n = 104)
------------------	---------------------------

2

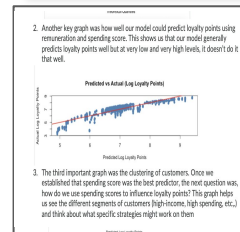
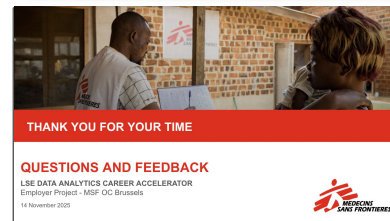
Interactive Power BI dashboard - depicts survey response % and all KPIs by project, country and by year.

3

PowerPoint Presentation - Provides high level insights, key takeaways and recommendations for MSF

4

Technical Report - Explains technical aspects of analysis, the thought process and analytical decisions taken to design dashboard

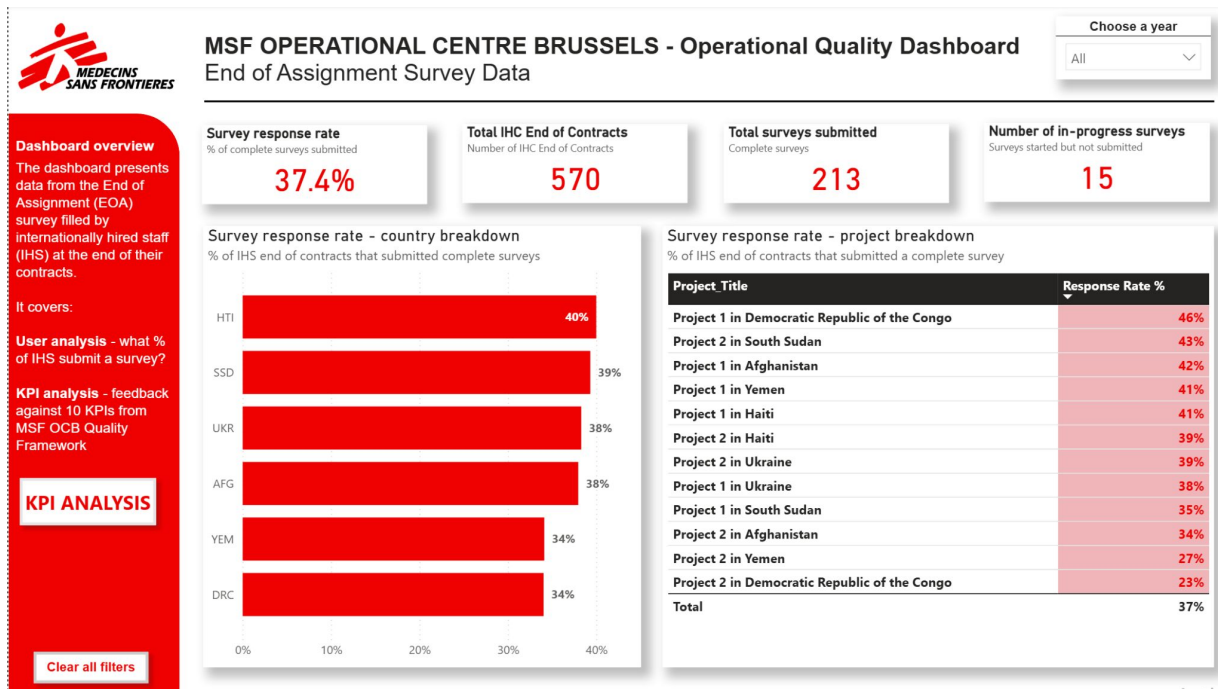


Recommendation 1: Improve the design of the EoA survey and enforce completion through the dashboard.

Insight: Survey response rates are critically low across all projects and countries.

We recommend:

- Flag low response countries and ensure Seniors drive completion.
- Introduce a mid-assignment survey to detect low-performing projects earlier.



Context

Findings

Recommendations

Recommendation 2: Create a holistic monitoring system to improve the delivery of projects.

Insight: *Projects show clear 'high' and 'low' performance against impact KPIs.*

We recommend:

- Produce case studies on performers.
- Turn lessons into practical guidelines for project design and delivery.

Insight: *Staff report major challenges around psychological support, safety, and conditions.*

We recommend:

- Flag repeated issues by country and address them proactively.
- Provide more psychological and emotional support in high-risk contexts.
- Improve safety procedures and living conditions for high-risk assignments.